

Exploring Scope Detection for Aspect-Based Sentiment Analysis

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Abstract—Aspect-based sentiment analysis (ABSA) aims to extract the aspect terms from review text, and to predict the polarity towards the aspect term. Although neural models have achieved competitive results, there are still many challenges in this task. Firstly, there is irrelevant and noise information in the review text, and the offsets of aspect term boundary are hard to decide. In addition, sentiment is usually either expressed implicitly or shifted due to the occurrence of negation and rhetorical words. To tackle the above limitations, we propose a scope detection model to distinguish whether the words from the review text are relevant with the aspect term, and to filter irrelevant and noise information. In addition, we investigate a biaffine-based model to constrain the scope detection process of aspect term extraction. We further generate a simplified clause based on the scope of aspect term, and predict the polarity based on the simplified clause. Empirical studies show the effectiveness of our proposed model over several strong baselines. These also justify the importance of scope detection for aspect-based sentiment analysis.

Index Terms—Aspect-based sentiment analysis, scope detection, text simplification, aspect term extraction, aspect-based sentiment classification.

I. INTRODUCTION

ASPECT-BASED sentiment analysis [1] aims to extract the aspect terms from review text, and to predict the polarity towards the aspect term. It can provide a fine-grained analysis of the users' opinions towards the specific aspect. Consequently, it has aroused much research attention in recent years [1], [2], [3], [4]. For example, in the sentence “The wine list is excellent”, the aspect term is “wine list”, and the polarity toward the aspect term is positive.

In this study, we focus on two of the most important sub-tasks in aspect-based sentiment analysis: aspect term extraction and aspect-based sentiment classification. Previous researches mostly regard *aspect term extraction* as a sequence labeling task and focus on developing various types of neural models [5], [6], [7], [8], [9] to generate a tag sequence for the review. Recently, span-based models have achieved success in aspect

Unfortunately, [the food here isn't anything less than exceptional], but everything else about this restaurant is the pits.



Fig. 1. Example of scope detection for aspect-based sentiment analysis.

term extraction [7], [10], [11], the key insight is to annotate each opinion target with its span boundary. Recent advances in *aspect-based sentiment classification* can be mainly categorized into several types: LSTM based [2], [12], memory based [13], [14], and pre-training methods [15], [16]. These models represent the sentence as a word sequence and neglect the syntactic relations between aspect and opinion words. To solve this problem, several recent researches [17], [18] leverage the graph-based models to incorporate the syntactic structure of a sentence and have shown better performance than those without considering syntactic relations.

Though achieving impressive progress, current methods still face several challenges. As shown in Fig. 1, there is irrelevant and noise information in the review text (e.g., the second sentence in the example). To tackle the above limitation, we propose a *scope detection model* to detect the scope of aspect terms. The *scope* refers to the extent or range of words in a review text that are related or relevant to a specific aspect term. This helps separate the important information from the unrelated parts of the text, such as “Unfortunately” and “the pits” in the example. The *scope detection model* can distinguish whether the words from the review text are relevant with the aspect term, and filter irrelevant and noise information. We then utilize a biaffine-based span detection model [19], [20] to detect the boundary of aspect term with biaffine-based constraints.

However, the opinion is still hard to capture, since it is usually either expressed implicitly or shifted due to the occurrence of negation and rhetorical words. In the Fig. 1 example, the presence of negation words like “isn’t anything less” and the word “exceptional” within the scope can still impact the neural model’s ability to predict sentiment polarity accurately. To overcome this obstacle, we further investigate the aspect-based

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simplification model to generate the simplified clause based on the scope of the aspect term. Notably, The simplified clause “the food here is good” is much easier to understand than the original review text. Afterward, we integrate the simplified clause with original review text for aspect-based sentiment classification.

We verify the effectiveness of the proposed model on both English and Chinese datasets. Experimental results show that our model outperforms competitive models, achieving the new state-of-the-art results on both benchmarks. In addition, the results also indicate the importance of scope detection for aspect-based sentiment analysis.

In summary, the main contributions of this article are concluded as follows:

- We design a novel aspect scope detection model to distinguish whether the words from the review text are relevant with the aspect term. It is also used to filter irrelevant and noise information.
- We employ biaffine-based constraints for scope detection of aspect term. To the best of our knowledge, it is the first time that an biaffine model is employed to constrain the scope detection of aspect term.
- We propose an aspect-based simplification model to generate the simplified clause based on the scope of the aspect term, and employ the simplified clause for aspect-based sentiment classification.
- We conduct extensive experiments and the results show that our proposed model outperforms the state-of-the-art methods. The results also indicate the importance of scope detection for aspect-based sentiment analysis.

II. RELATED WORK

In this section, we introduce the related works on two subtasks of aspect-based sentiment analysis: aspect term extraction and aspect-based sentiment analysis.

A. Aspect Term Extraction

Early researches for aspect term extraction mainly involve pre-defined rules [1], [21], [22] and hand-craft features [23], [24]. With the development of deep learning techniques, neural methods have become the mainstream. Aspect term extraction can be viewed as either a supervised or an unsupervised task. For unsupervised aspect term extraction, the commonly used neural methods are based on topic models [25] or attention mechanism [26], or conditional random fields [27], [28]. For supervised aspect term extraction, the researchers focus on developing various types of neural sequence taggers [5], [6], [7], [8]

Recently, span-based models has achieved successful in aspect term extraction [7], [10], [11], the key insight is to annotate each opinion target with its span boundary. For example, Hu et al. [10] proposed a span-based extract-then-classify framework, where multiple opinion targets were directly extracted from the sentence under the supervision of target span boundaries, and corresponding polarities were then classified using their span representations. Meanwhile, Mao et al. [11] employed

machine reading comprehension mechanisms with parameters sharing to extract aspect terms as well as opinion words.

Except extracting aspect term individually, a recent trend on aspect-based sentiment analysis is towards the unified framework [10], [29], [30], [31], where the interactive relations between aspect term extraction, opinion term extraction, or aspect-level sentiment classification are exploited to enhance the overall performance. For example, Yu et al. [32] proposed a multi-task learning framework to implicitly capture the relations between aspect term extraction and opinion term extraction tasks. Similarly, Wang et al. [33] proposed a joint Recursive Neural Conditional Random Fields (RNCRF) model by using the dependency parse tree to capture dual-propagation among aspect terms and opinion words. Xu et al. [34] trained BERT on domain-specific data to boost its sequence labeling performance. Li et al. [35] proposed to generate additional datasets for improving the performance of ATE. Zhao et al. [9] treated the problem from a perspective of joint term and relation extraction rather than under the sequence tagging formulation. They proposed a multi-task learning framework based on shared spans, where the terms were extracted under the supervision of span boundaries.

B. Aspect-Based Sentiment Classification

The goal of aspect based sentiment classification is to identify the sentiment polarity of an aspect within a given context sentence. The recent studies focus on developing various types of deep learning models. We briefly review the neural models without considering syntax, and then go to the syntax based ones.

Recent advances focus on developing various types of neural models. A majority of the work used LSTM neural networks to model the word sequence in a sentence [2], [12]. Except for these LSTM-based methods, there are some other neural methods existing in the literature, including convolutional neural networks [36], [37] and deep memory networks [13], [14]. Benefited from the rich linguistic knowledge learned from massive language modeling [38], researchers show great progress on this task using BERT representations [15], [16], [34].

Some other efforts try to directly include the syntactic information. Since aspects are generally assumed to lie at the heart of this task, establishing the syntactic connections between each target aspect and the other words are crucial. Qiu et al. [22] manually defined some syntactic rules to identify the relations between aspects and potential opinion words. Afterward, neural network models were explored for this task. Dong et al. [39] and Nguyen and Shirai [40] combined the dependency and constituent trees of a sentence with a recursive neural network for aspect-based sentiment classification.

Recently, graph neural networks combined with dependency trees have shown appealing effectiveness in aspect-based sentiment classification [3], [41], [42], [43]. The basic idea is to transform the dependency tree into a graph, and then impose the graph neural networks to propagate information from syntax neighborhood opinion words to aspect words. Sun et al. [41] and Zhang et al. [17] used graph convolutional networks to learn node representations from a dependency tree and used them

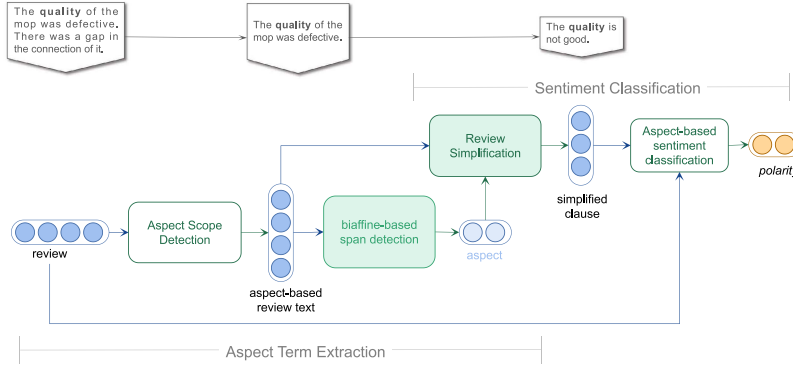


Fig. 2. Overview of proposed model.

together with other features for sentiment classification. For a similar purpose, Huang et al. [42] and Wang et al. [18] used graph attention networks to explicitly establish the dependency relationships between words. Liang et al. [44] employs a graph attention network to fully exploit the syntax information of the constituent tree of a sentence.

Much more recently, unified generative architecture has shown its dominance in aspect-based sentiment analysis [45], [46], [47], [48]. These methods usually employ generative pre-trained language models to convert the original task into a text generation problem, require to generate quad-tuples for aspect-based sentiment analysis in an end-to-end manner. For example, Gao et al. [46] proposes a unified generative framework for aspect-based sentiment analysis, by controlling the type of task prompts consisting of multiple element prompts.

Different from previous studies, which focus on the original review text directly, we consider the importance of scope of aspect terms, and extract the aspect terms and detect the polarity based on the scope of aspect terms. The scope of aspect terms can be used to distinguish whether the words from the review text are relevant with the aspect term, and filter irrelevant and noise information.

It's important to emphasize that our objective is not review summarization or simplification in the traditional sense. Our focus is on sentiment classification through a simplification approach. In this context, text simplification aims to make the text more understandable to individuals with lower reading abilities [49]. Our specific method involves the removal of negation and rhetorical expressions to present sentiment in a clearer manner. To the best of our knowledge, we are the first to introduce a text simplification model into aspect-based sentiment classification.

III. SCOPE DETECTION FOR ASPECT-BASED SENTIMENT ANALYSIS

Fig. 2 gives a work-flow of proposed model. We firstly employ a scope detection model to detect the scope of the aspect term. Secondly, we use a biaffine-based span detection model to constrain the process of aspect term extraction. Thirdly, we utilize an aspect-based simplification model to generate the simplified clause based on the scope of aspect terms. Afterward,

we integrate the simplified clause with original review text for aspect-based sentiment classification. In the following sections, we will discuss these issues one by one.

A. Review Representation

Firstly, given an aspect term, we employ BERT [38] to learn the representation of word sequence $W = \{w_1, w_2, \dots, w_n\}$. Then, we convert each token $w_i \in W$ into vector space by summing the token, segment, and position embeddings. We use a series of L stacked TransformerBlock to project the input embeddings into a sequence of contextual vectors H_W^i as:

$$H_W^i = \text{TransformerBlock}(H_W^{i-1}), \quad \forall i \in [1, L] \quad (1)$$

In this way, we obtain the representation $H = H_W^L$ for the review text.

B. Aspect Scope Detection

Aspect scope detection aims to detect the start/end positions of the scope of aspect term. As shown in Fig. 1, although the original review text is very long, only the first sentence is correlated with aspect term, and can be selected as the scope of aspect term. Therefore, aspect scope detection helps distinguish whether the words from the review text are relevant with the aspect term and can be used to filter the irrelevant and noise information from the review text. Meanwhile, its score can be regarded as an indicator of coherence during inference, helping to select the most coherent aspect term from multiple candidate aspect terms.

As shown in Fig. 3, given a sequence $H = \{h_1, h_2, \dots, h_n\}$ from review representation, we apply two separate FFNN to create different representations (h_s/h_e) for the start/end of the spans. Using different representations for the start/end of the spans allow the system to identify the start/end of the spans separately.

$$h_s(i) = \text{FFNN}_s(h_i) \quad (2)$$

$$h_e(i) = \text{FFNN}_e(h_i) \quad (3)$$

We then introduce two new parameter matrices $W_s \in R^{H \times T}$ and $W_e \in R^{H \times T}$, and apply sigmoid to produce the probability

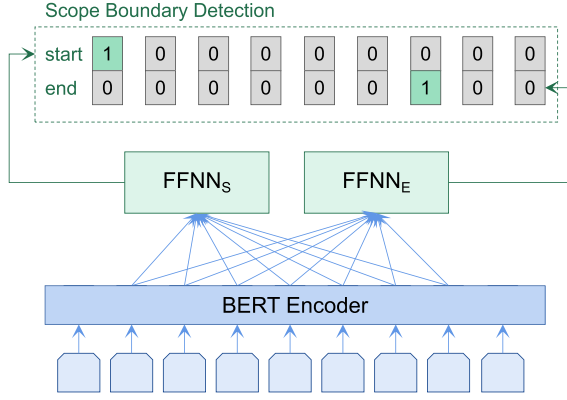


Fig. 3. Aspect scope detection model.

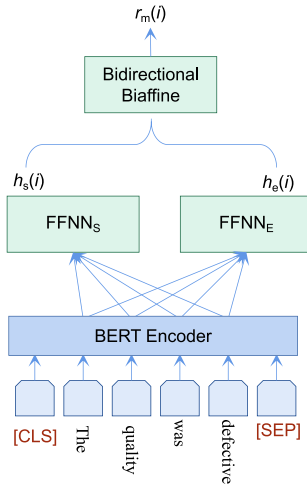


Fig. 4. Aspect term extraction with biaffine constraints.

of each token being selected as the start/end of scope:

$$S_s(i) = \text{sigmoid}(h_s(i)W_s) \quad (4)$$

$$S_e(i) = \text{sigmoid}(h_e(i)W_e) \quad (5)$$

where $S_s(i)$ and $S_e(i)$ is the output of the scope detection model, and is used to predict the start and end offsets of the scope of aspect term.

C. Aspect Term Extraction With Biaffine Constraints

As shown in Fig. 4, we reformulate aspect term extraction as the task of identifying start and end indices. In addition, we employ a bidirectional biaffine constraints model to assign scores to all possible spans in a sentence. We believe that the bidirectional biaffine constraints can learn information from two directions, and improve the robustness of span detection. After that, we rank the candidate spans by their scores and return the top-ranked spans that comply with constraints as aspect terms.

In this subsection, we firstly give a general description of span-based aspect term extraction, and we then introduce the biaffine-based constraints.

Span-based Aspect Term Extraction: In this study, we employ a span-based model for aspect term extraction. Firstly, we use the hidden states and the probability of offset from the scope

detection model as the input,

$$t_s(i) = h_s(i) \oplus S_s(i) \quad (6)$$

$$t_e(i) = h_e(i) \oplus S_e(i) \quad (7)$$

and we employ two separate FFNN to create different representations (b_s/b_e) for the start/end of the spans,

$$b_s(i) = \text{FFNN}_s(t_s(i)) \quad (8)$$

$$b_e(i) = \text{FFNN}_e(t_e(i)) \quad (9)$$

We then apply sigmoid to produce the probability of each token being selected as the start/end of the span:

$$T_s(i) = \text{sigmoid}(b_s(i)W_s) \quad (10)$$

$$T_e(i) = \text{sigmoid}(b_e(i)W_e) \quad (11)$$

where $T_s(i)$ and $T_e(i)$ denote the start/end offset of aspect term.

Bidirectional Biaffine Constraints: After we employ the span-based model to extract aspect terms, we employ a bidirectional biaffine model to constrain the boundary extraction process of aspect terms. In particular, the bidirectional biaffine model is used to over the sentence to create $l \times l \times 1$ scoring tensors (r_s/r_e), where l is the length of the sentence,

$$r_s(i) = B_s(i)^T U h_e(i) + W(B_s(i) \oplus B_e(i)) + b \quad (12)$$

$$r_e(i) = B_e(i)^T U h_s(i) + W(B_e(i) \oplus B_s(i)) + b \quad (13)$$

where $B_s(i) = b_s(i) \oplus T_s(i)$, $B_e(i) = b_e(i) \oplus T_e(i)$, s_i and e_i are the start and end indices of the span i , U is a $d \times c \times d$ tensor, W is a $2d \times c$ matrix and b_m is the bias.

The tensor r_s and r_e provides scores for all possible spans that could constitute an aspect term under the constrain that $s_i \leq e_i$ (the start of entity is before its end). Therefore, we assign each span a category:

$$s_s(i) = \text{argmax}(T_s(i) \times r_s(i)) \quad (14)$$

$$s_e(i) = \text{argmax}(T_e(i) \times r_e(i)) \quad (15)$$

We then rank all the spans that is an aspect term or not in descending order and apply following post-processing constraints: a term is selected as long as it does not clash the boundaries of higher ranked entities. We denote a term i to clash boundaries with another term j if $s_i < s_j \leq e_i < e_j$ or $s_j < s_i \leq e_j < e_i$. Hence only the span with higher score will be selected as aspect term.

D. Aspect-Based Sentiment Classification With Review Simplification

After we extract the aspect terms and detect the scope of aspect terms, we employ a review simplification model to generate the simplified clause, and use the simplified clause for aspect-based sentiment classification.

In our aspect-based review simplification model, there are three elements: aspect-based review text r , simplified clause c , and latent variables from aspect-based sentiment analysis (i.e., z_t

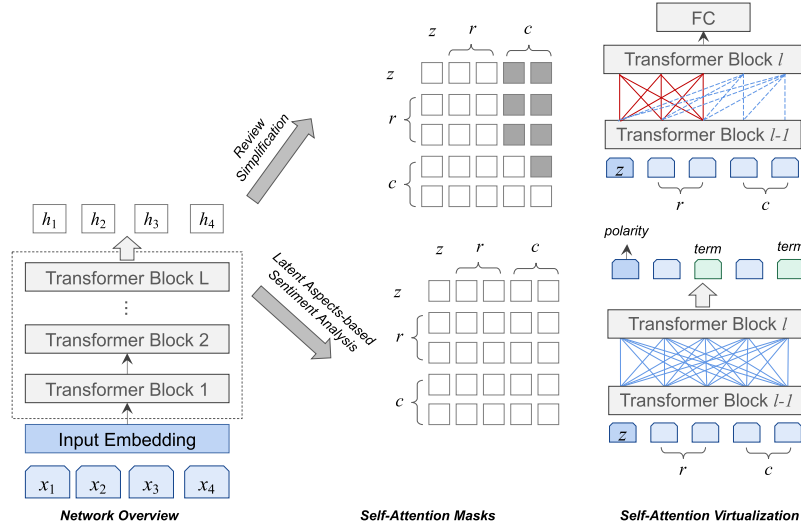


Fig. 5. Architecture of aspect-based review simplification model.

for aspect term and latent variable z_o for opinion). The simplified clause c is used to simplify the aspect-based review text, it is one piece of appropriate clause towards the given review text t . The latent variable z_t and z_o are used to make sure the simplified clause c contains the aspect term, and shares the same opinion with the original review text. Therefore, our aspect-based review simplification model contains the following two tasks – review simplification and latent aspect-based sentiment analysis.

We propose a unified infrastructure for the joint learning of all the tasks, shown as Fig. 5. The backbone of our infrastructure supports both bi-directional encoding and uni-directional decoding flexibly via specific self-attention masks [50], [51]. The review simplification and latent aspect-based sentiment analysis are carried out under the unified network with shared parameters.

Review Simplification: The review simplification model aims to simplify the aspect-based review text to a simplified clause. Since the pre-trained language models have been employed as encoders for natural language understanding problems [50], [52], we employ the pre-trained language model to generate the simplified clause from aspect-based review text.

As shown in Fig. 5, given the aspect-based review text r , a specific aspect term z_t , and the opinion z_o , the simplified clause can be estimated as,

$$p(c | r, z_t, z_o) = \prod_{t=1}^T p(c_t | r, z_t, z_o, c_{<t}) \quad (16)$$

where T is the length of the target simplified clause c , and $c_{<t}$ denotes previously generated words. Since the review simplification is a uni-directional decoding process, each token in the clause only attends to those before it.

Latent Aspect-based Sentiment Analysis: The latent aspect-based sentiment analysis task is included to identify the corresponding value of aspect term z_t and opinion z_o for the given aspect-based review text and simplified clause in the training data. The latent tasks shares network parameters with

review simplification, but has a separate self-attention mask for bi-directional encoding.

As shown in Fig. 5, with a special mask symbol $[M]$ as input, it keeps collecting information from the review context and target simplified clause. In this way, the corresponding aspect term and opinion for the target clause can be recognized as $z_t \sim p(z_t | c, r)$ and $z_o \sim p(z_o | c, r)$, where $p(z_t | c, r)$ and $p(z_o | c, r)$ are the estimated posterior distribution over discrete latent values.

The posterior distribution over latent values of z_t is estimated through the task of latent aspect term extraction:

$$p(z_t | c, r) = \frac{\exp(\text{score}(z_t, c))}{\sum_{z'_t} \exp(\text{score}(z'_t, c))} \quad (17)$$

where $\text{score}(z, c)$ is a CRF-based score function that is defined as the summation of transition scores and emission scores from representation of aspect-based review text.

Meanwhile, the posterior distribution over latent values of z_o is estimated through the task of latent opinion polarity prediction:

$$p(z_o | c, r) = \text{softmax}(W_2 h_{[M]} + b_2) \quad (18)$$

where $h_{[M]} \in \mathbb{R}^D$ is the final hidden state of the special mask, $W_2 \in \mathbb{R}^{K \times D}$ and $b_2 \in \mathbb{R}^K$ denote the weight matrices of one fully-connected layer.

Aspect-based Sentiment Classification: After we obtain the simplified clause, we integrate it with original text for aspect-based sentiment classification. Given the hidden representation from original text H and simplified clause H_c , we employ a multi-layer perceptron to predict polarity by concatenating $\{H, H_c\}$ as follow:

$$H_P = \sigma(W_p^h (H \oplus H_c) + b_p^h), \quad (19)$$

H_P is used as inputs to a softmax output layer:

$$P_P = \text{softmax}(W_p H_P + B_P) \quad (20)$$

Here, W_p^h , b_p^h , W_p , and B_p are model parameters, and P_P is used to predict polarity from original text and simplified clause jointly.

E. Training

We train the aspect-based sentiment classification model with the review simplification model and the aspect-based scope detection model in a unified architecture.

Loss of aspect-based scope detection: To train the aspect-based scope detection model, we minimize the negative log-likelihood loss, parameters are updated during the training process. In particular, the loss is the sum of two parts: the start token loss and end token loss,

$$\mathcal{L}_S = - \sum_i y_i^s \log(p_i^s) - \sum_i y_i^e \log(p_i^e) \quad (21)$$

where y_i^s and y_i^e are ground truth start and end positions for the scope detection model.

Loss of aspect term detection: The loss of aspect term detection is the sum of two parts: the start token loss and end end token loss,

$$\mathcal{L}_T = - \sum_i y_i^s \log(T_i^s) - \sum_i y_i^e \log(T_i^e) \quad (22)$$

where y_i^s and y_i^e are ground truth start and end positions for the aspect term detection model.

Loss of aspect-based review simplification: The simplified clause is generated conditioned on the latent variable and the context of original review text. Therefore, we employ negative log-likelihood loss,

$$\mathcal{L}_C = -\mathbb{E}_{z_t \sim p(z_t|c,r), z_o \sim p(z_o|c,r)} \log(p(c | r, z_t, z_o)) \quad (23)$$

where z_t and z_o is the latent aspect term and opinion of this training pair (t, c) , sampled from the probability distribution $p(z_t | t, c)$ and $p(z_o | t, c)$.

Loss of aspect-based sentiment classification: Our training objective of aspect-based sentiment classification is to minimize the cross-entropy loss over a set of training examples $(d_i, y_i)_{i=1}^N$, with a ℓ_2 -regularization term,

$$\mathcal{L}_P = - \sum_{i=1}^N \sum_{j=1}^K y_i \log \hat{y}_i + \frac{\lambda}{2} \|\theta_y\|^2 \quad (24)$$

where y_i is the pre-defined polarity, $\hat{y}_i$ is the predicted label, θ_y is the set of model parameters and λ is a parameter for ℓ_2 -regularization.

Therefore, the final loss is,

$$\mathcal{L} = \lambda_1 \mathcal{L}_S + \lambda_2 \mathcal{L}_T + \lambda_3 \mathcal{L}_C + \lambda_4 \mathcal{L}_P \quad (25)$$

where λ are the trainable weight parameters, and $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$.

IV. EXPERIMENTATION

In this section, we introduce the datasets used for evaluation and the baseline methods employed for comparison. We then

TABLE I
STATISTICS OF DATASETS

Dataset	Sentences		Aspects	
	Train	Test	Train	Test
Rest	1,266	492	2,051	848
Laptop	906	328	1,281	463
Chan	5,152	1,718	12,266	4,209
Furn	5,587	1,863	19,695	6,506
Kith	5,313	1,771	13,343	4,531

report the experimental results conducted from different perspectives, and analyse the effectiveness of proposed model with different factors.

A. Data and Annotation

1) *Dataset:* We evaluated our proposed model on both English and Chinese dataset. The English dataset consists of reviews on restaurant domain (Rest) and laptop domain (Laptop) from ASTEDData-V2.¹ ASTEDData-V2 [31] is refined dataset from SemEval-14 tasks [53], it removes the samples with conflicting polarities and those without explicit aspects in the sentences. The Chinese datasets is collected from Taobao.com, one of the top online shopping websites in China. There are three domains in the dataset, i.e., Chandlery (Chan), Furniture (Furn), Kitchen (Kith).

The statistics of all the datasets are shown in Table I. From the table, we find that there are more than one aspect term in each review. Since the scope of each aspect term is different, it suggests us to detect the scope for aspect term extraction.

2) *Annotation:* The systematic extraction of ground truth aspect scopes adheres to three straightforward rules: (1) The scope should be the most compact phrase possible, delimited by punctuation marks. (2) The scope must encompass both an aspect word and an accompanying adjective or adverb, with sentiment determined based on the HowNet sentiment lexicons [54]. (3) The scope's length should not exceed 10 words.

The ground truth simplified clause are annotated manually from the original text. In the annotation process, five annotators, all possessing bachelor's degrees and strong proficiency in both Chinese and English, were engaged. The initial step involved all five annotators independently annotating "Trivial Reviews," comprising 100 English and 100 Chinese reviews, followed by regular meetings to establish annotation standards. Subsequently, four annotators individually annotated all the remaining reviews. Whenever disagreements arose, the fifth annotator stepped in, and annotations were determined through a majority vote. The scope annotation rules are as follows:

- Both aspect term and opinion word should be in the simplified clause.
- The polarity of the simplified clause should be the same as the original review text.
- The simplified clause should be as shorter as it can.

We then give the statistic of length between original review text and simplified clause in Table II. From the table, we can find that although these simplified clauses consist of aspect terms and

¹[Online]. Available: <https://github.com/xuuluuuu/SemEval-Triplet-data>

TABLE II
LENGTH OF ORIGINAL REVIEW TEXT AND SIMPLIFIED CLAUSE

Dataset	Original Text	Simplified Clause
Rest	32.5	10.5
Chan	64.1	10.3
Furn	103.4	11.6
Kith	75.3	11.9

opinion words, they are still much short than the original text. It may due to that there is a lot of noise and irrelevant information in reviews. Therefore, it indicates that the scope of aspect terms are very important for aspect-based sentiment analysis, and also suggests us to use the simplified clause instead of the original text for aspect-based sentiment classification.

B. Implementation

We use the case BERT_{base} version of BERT and fine-tune its parameters during training in this work. We tune the parameters of our models by grid searching on the validation dataset. We select the best models by early stopping using the F1-score results on the validation dataset. The dimension of other hidden variables of all the models is 128. All The model parameters, including λ in Equation 25, are optimized by Adam [55] with a learning rate of 2e-5. The batch size is 32 and a dropout probability of 0.1 is used.

The experimental results are obtained by averaging ten runs with random initialization. We use F1-score to evaluate the performance of aspect term extraction. We consider a predicted aspect term is correct if the gold standard annotations contain a aspect term the same as the prediction. Meanwhile, we employ Accuracy as the evaluation metrics for aspect-based sentiment classification.

C. Baseline Models for Experimental Comparison

1) Only for Aspect Term Extraction:

- *BiLSTM+CRF* is a sequence tagging method with a BiLSTM network built on top of pre-trained word embeddings, followed by a CRF output layer to perform BIO classification.
- *BERT+CRF* is a sequence tagging method based on a BERT encoder. The output hidden states of input words are taken as the features for CRF.
- *CMLA* [56] is a joint model with multi-layer attentions for aspect term and opinion word extraction.
- *SPAN* [10] is a pipeline method using BERT as the backbone, and employs a span-based multi-target extractor for aspect term extraction.
- *BiSELF-CRF* [7] employs a pointer network to reposition the boundaries for aspect term extraction.

2) Only for Aspect-Based Sentiment Classification:

- *ATAE-LSTM* [57] is a classic LSTM based model which explores the connection between an aspect and the content of a sentence with an attention-based LSTM.
- *MemNet* [13] is a memory based method combining a neural attention model with an external memory to calculate the importance of each context word towards an aspect.

TABLE III
RESULTS OF ASPECT TERM EXTRACTION

Method	Rest	Laptop	Chan	Furn	Kith	Average
BiLSTM+CRF	70.0	68.2	54.8	52.3	53.9	59.9
BERT+CRF	77.7	77.2	64.9	64.2	66.9	70.2
CMLA	79.6	80.3	66.8	64.9	67.7	71.9
SPAN	78.7	78.0	66.8	68.0	67.1	71.8
BiSELF-CRF	79.4	79.9	68.3	66.4	68.9	72.6
GAS	79.7	80.4	67.2	66.5	67.7	72.3
LEGO-ABSA	81.0	81.4	69.2	67.9	70.3	74.0
OTG	79.3	80.7	68.5	67.4	68.3	72.9
Ours	81.3	81.6	70.5	69.4	72.5	75.0

- *TD-BERT* employs BERT [38] to model the preceding and following contexts surrounding the aspect terms for aspect-based sentiment classification [2].
- *R-GAT* [18] employs a relational graph attention to encode the aspect-oriented dependency tree structure network sentiment prediction.
- *METNet* [16] is a mutual enhanced transformation network. It employs an aspect enhancement module and hierarchical structure to improves the representation learning for aspect-based sentiment classification.

3) For Both Sub-Tasks:

- *GAS* [47] uses a pre-trained language model to perform unified aspect-based sentiment analysis, generating aspect terms and their polarity pairs jointly in either an annotation or extraction style.
- *LEGO-ABSA* [46] proposes a unified generative multi-task framework that can solve multiple ABSA tasks by controlling the type of task prompts consisting of multiple element prompts.
- *OTG* [48] uses a generative model to generate a tree structure that contains all aspect-level sentiment quadruplets which correspond to the aspect category, aspect term, opinion term, and sentiment polarity, respectively.

D. Results of Aspect Term Extraction

In this subsection, we give the results of aspect term extraction, and analyze the influence of the factors in the proposed model.

1) *Comparison With Baselines:* We compare the proposed model with various strong baselines on Table III. From the table, we find that our proposed model achieves the best results in terms of macro-F1 on all datasets. The results show that the proposed model outperforms both sequence tagging-based and span-based models.

By contrast, span-based models (i.e., SPAN, BiSELF-CRF) are strong baselines by reducing the search space with a multi-target extractor. Therefore, span-based models outperform sequence tagging models in most domains.

Another observation is that generative methods(i.e., GAS, LEGO-ABSA, OTG) generally outperforms the sequence-tagging models and span-based models. However, GAS and OTG perform relatively worse compared to LEGO-ABSA and our model, possibly because they are optimized for unified aspect analysis and aspect quadruple extraction (term-category-sentiment-opinion), respectively, leading to subpar performance

TABLE IV
INFLUENCE OF DIFFERENT FACTORS IN ASPECT-BASED TERM EXTRACTION MODEL

Domain	SPAN	+Scope	+BiBC	Ours
Rest	78.7	80.8	82.1	81.3
Laptop	78.0	79.1	81.6	81.6
Chan	66.8	68.5	70.0	70.5
Furn	68.0	68.9	69.3	69.4
Kith	67.1	71.2	71.7	72.5
Average	71.8	73.7	74.9	75.0

TABLE V
RESULTS OF ASPECT-BASED SENTIMENT CLASSIFICATION

Method	Rest	Chan	Furn	Kith
ATAE-LSTM	74.4	83.3	82.8	85.8
MemNet	73.9	85.0	83.0	85.0
TD-BERT	79.2	84.8	87.6	87.5
R-GAT	82.5	83.8	87.5	88.3
METNet	82.0	83.3	86.8	86.3
GAS	82.3	84.2	87.1	87.9
LEGO-ABSA	83.3	86.2	88.6	89.9
OTG	82.9	83.6	87.7	88.2
Ours	83.7	87.5	90.8	90.8

when predicting only the term. Nevertheless, our model still outperforms all span-based models and competes well with recent generative methods. It shows the importance of both aspect scope detection and biaffine constraints for aspect term extraction.

2) *Influence of Different Factors*: We then analyze the influence of different factors of the proposed model. As shown in Table IV, we employ the basic span-based model SPAN as the baseline method. “+Scope” means employs the aspect scope boundary detection model to distinguish whether the words from the review text are relevant with the aspect term. “+BiBC” means employs a bidirectional biaffine-based span detection model to extract aspect terms with biaffine-based constraints.

From the table, we can find that the aspect scope detection model outperform the basic SPAN model significantly. It shows that the boundary of aspect scope is very effective for extracting aspect terms. Meanwhile, we find that “+BiBC” is more effective than “+Scope”, which indicates that biaffine-based constraints are more important for detecting the boundary of aspect span.

In addition, our proposed model outperforms all the other models. It indicates the effectiveness of the proposed model, and also suggests that we need to integrate both aspect scope detection and biaffine constraints for aspect term extraction.

E. Results of Aspect-Based Sentiment Classification

In this subsection, we show the results of aspect-based sentiment classification, and we also analyze the impact of different factors in the proposed model.

1) *Comparison With Baselines*: We compare the proposed model with various strong baselines on Table V. From the table, we can find that our proposed model achieves the best results in terms of accuracy on all datasets. It shows the effectiveness of the proposed model for aspect-based sentiment classification.

By contrast, although R-GAT employs a graph neural model to capture the dependency relations, and METNet employs a

TABLE VI
INFLUENCE OF DIFFERENT FACTORS IN ASPECT-BASED SENTIMENT CLASSIFICATION

Method	Rest	Chan	Furn	Kith
TD-BERT	79.2	84.8	87.6	87.5
+Simp	80.4	86.2	84.8	88.3
+Scope	79.0	86.2	87.3	86.7
+Simp+Scope	80.7	86.5	87.5	89.1
+Simp+Sentiment	82.2	86.3	86.8	88.5
Ours	83.7	87.5	90.8	90.8

hierarchical structure for representation learning, our proposed model still outperforms them significantly. Similar to the results in Table III for aspect-term extraction, generative baselines(i.e., GAS, LEGO-ABSA and OTG) performs better generally than other baselines, but the methods designed for pair extraction or quadruple extraction performed worse than our models significantly. Still, our model give the best performance in this task. It indicates the effectiveness of the proposed model and shows the importance of simplified clause for aspect-based sentiment classification.

2) *Influence of Different Factors*: We then analyze the effect of scope-detection in aspect-based sentiment classification. As shown in Table VI, we employ TD-BERT as the baseline method. “+Simp” only employs the review simplification model to generate the simplified clause. “+Scope” employs the aspect-based scope detection model to detect the scope of aspect terms. “+Sentiment” employs latent aspect-based sentiment analysis as an auxiliary task for simplified clause generation.

From the Table VI, we can find that both review simplification (“+Simp”) and scope detection (“+Scope”) outperforms the basic TD-BERT model significantly. It shows that the simplified clause is very effective for aspect-based sentiment classification. Meanwhile, it also indicates that the scope of aspect term is very helpful. We also find that “+Simp+Sentiment” outperforms “+Simp”, it shows that the latent aspect-based sentiment analysis is very important for generating simplified clause.

In addition, our proposed model outperforms all the other models. It shows the effectiveness of the proposed model. It also suggests that we should integrate both aspect-based scope detection and review simplification for aspect-based sentiment classification.

V. ANALYSIS AND DISCUSSION

In this section, we give some analysis to discuss the importance of scope detection, give some error analysis and discussion are conducted with a few representative examples.

A. Analysis of Scope Detection Methods

Firstly, we analyze the effectiveness of different scope detection methods for extracting the scope of aspect term in reviews. As shown in Table VII, “Rule” uses rule-based method to detect the boundary of scope with punctuations and make sure that aspect terms are in the clause. “CRFs” employs a neat and effective multiple CRFs based structured attention model to extract the boundary of scope for aspect term extraction [58].

TABLE VII
EFFECTIVENESS OF DIFFERENT SCOPE DETECTION METHODS

Domain	SPAN	Rule	CRFs	Ours
Rest	78.7	76.9	76.2	81.3
Laptop	78.0	77.2	76.9	81.6
Chan	66.8	67.2	68.7	70.5
Furn	68.0	67.8	68.2	69.4
Kith	67.1	67.8	70.8	72.5
Average	71.8	71.4	72.2	75.0

TABLE VIII
ANALYSIS OF BIAFFINE CONSTRAINTS

Domain	SPAN	+BC	+BiBC	Ours
Rest	78.7	80.9	82.1	81.3
Laptop	78.0	80.3	81.6	81.6
Chan	66.8	69.8	70.0	70.5
Furn	68.0	68.6	69.3	69.4
Kith	67.1	70.1	71.7	72.5
Average	71.8	73.9	74.9	75.0

From the results, we find that both the rule-based and CRF-based model are not very effective. Since the aspect terms are not given, it is hard for them to distinguish relevant words with the aspect term. In addition, the proposed model outperforms all other models significantly, it shows the effectiveness of the proposed model, and it also indicates that the importance of the scope for aspect term extraction.

B. Analysis of Biaffine Constraints

In this subsection, we analyze the effectiveness of different biaffine constraints for aspect term extraction. As shown in Table VIII, we employ the basic span-based model SPAN as the baseline method. “+BC” means employs a biaffine-based span detection model [20] to extract aspect terms with biaffine-based constraints. “+BiBC” means employs the proposed bidirectional biaffine-based span detection model to extract aspect terms with biaffine-based constraints.

From the table, we can find that all the biaffine-based span detection models outperform the basic SPAN model. It indicates the importance of biaffine constraints for aspect term extraction. Meanwhile, the proposed bidirectional biaffine constraints (“+BiBC”) outperform the traditional biaffine constraints in all the domains, which shows the effectiveness of the proposed bidirectional biaffine constraints for aspect term extraction.

Our proposed model exhibits superior overall performance compared to other models. However, an intriguing observation in the table is the remarkable performance of “+BiBC” on the English dataset, even surpassing our proposed model’s performance on the restaurant dataset. It may be due to the fact that “+BiBC” relies on dependency parsing [20], and Chinese dependency parsing is inherently more complex and challenging than that of English [59]. Consequently, the quality of dependency parsing for Chinese may be inferior to that of English, potentially explaining the performance of “+BiBC” on English data, especially on Restaurant. While “+BiBC” stands as a very strong baseline, it encounters difficulties when confronted with Chinese datasets. On the other hand, our model achieves a superior overall performance by integrating biaffine constraints

TABLE IX
ANALYSIS OF DIFFERENT GENERATION METHODS

Method	Rest	Chan	Furn	Kith
Rule	81.0	84.9	88.8	88.2
TD-BERT	79.2	84.8	87.6	87.5
+Simp	80.4	86.2	84.8	88.3
+Simp+Aspect	81.5	86.3	88.6	87.8
+Simp+Opinion	83.1	86.8	88.1	88.8
Ours	83.7	87.5	90.8	90.8

TABLE X
IMPORTANCE OF SIMPLIFIED CLAUSE WITH DIFFERENT CLASSIFICATION MODELS

Method	Rest	Chan	Furn	Kith
TD-BERT	79.2	84.8	87.6	87.5
TC-Clause	60.2	64.3	67.0	68.5
LSTM-Clause	64.5	70.8	72.3	74.0
BERT-Clause	73.1	85.2	81.3	88.1
Ours	83.7	87.5	90.8	90.8

TABLE XI
EXAMPLES OF CASE STUDY ON ASPECT TERM EXTRACTION

ID	Review	SPAN	Ours
1	The packaging is exquisite, the quality is very good, the silicone material is cold, I am very satisfied on it.	material	silicone material
2	First of all, the express package of this pot is very good. There are three layers inside and outside. The pot cover has both hand feel and texture. The pot with sharp bottom is easy to use.	None	package, pot

and aspect scope detection, only with an acceptable trade-off in performance on the Restaurant dataset.

In the end, the experimental results indicate the effectiveness of the proposed model, and also suggests that we need to integrate both aspect scope detection and biaffine constraints for aspect term extraction.

C. Analysis of Different Simplified Clause Generation Methods

We then analyze the effectiveness of different generation methods for generating the simplified clause in Table IX, where “Rule” means use rule-based method to detect the boundary of the clause with punctuations and make sure that aspect terms are in the clause. “+Aspect” and “+Opinion” means consider latent aspect-based sentiment analysis as an auxiliary task and extract aspect term or opinion for simplified clause generation respectively.

From the results, we find that even a rule-based model outperforms the basic TD-BERT. It shows that the scope of aspect terms is very important for aspect-based sentiment classification. However, some expressions may be still hard to understanding in the scope of aspect terms. Therefore, “+Simp” outperforms the rule-based model. It shows that “+Simp” can generate an easier understanding clause than the rule-based model.

In addition, we find that both “+Simp+Aspect” and “+Simp+Sentiment” outperforms “+Simp”, it shows that both

TABLE XII
EXAMPLES OF CASE STUDY ON ASPECT-BASED SENTIMENT CLASSIFICATION

ID	Original Review Text	Simplified Clause	TD-BERT	Ours
1	Other said that the pot is very good, and I need a new pot, so I bought a 32cm uncovered pot. After use, I find that the pan fried dumplings are very sticky. I hope it can be changed. The appearance and details of this pot are good.	pan is sticky.	+1	-1
2	Other said that the pot is very good, I've been waiting for delivery for 20 days. I've been told that there are too many people to buy, so delivery is slow. But just the order number shows the status of picking up items, which shows ten days of things. I really want to say that the customer service attitude is OK. Overall, it didn't disappoint me.	the delivery is too show.	+1	-1

aspect term and opinion words are very effective for generating simplified clause.

D. Importance of Simplified Clause

After analyzing the effectiveness of different generation methods, we then analyze the importance of simplified clauses with different classification models. As shown in Table X, “TC-Clause” employs a term-counting method to predict the polarity by counting the number of positive and negative words in a simplified clause. “LSTM-Clause” and “BERT-Clause” integrate the simplified clause into the classification model to predict the polarity. Note that, different from the proposed model (Ours) which integrates both original text and simplified clause, “TC-Clause”, “LSTM-Clause” and “BERT-Clause” only utilize the simplified clause for sentiment classification.

From the results, we find that the weak classification models “TC-Clause” and “LSTM-Clause” can reach an acceptable result. It shows that the effectiveness of simplified clauses for aspect-based sentiment classification.

In addition, although “BERT-Clause” only employs the simplified clause for aspect-based sentiment classification, it still reaches a comparable result with TD-BERT. It shows the simplified clause is very important and useful for aspect-based sentiment classification.

Finally, our proposed model outperforms “BERT-Clause” significantly, it also suggests that we need to integrate both original text and simplified clause for sentiment classification.

E. Case Study

In this subsection, we give some case studies to show the effectiveness of proposed model, and analyze the importance of scope detection for aspect-based sentiment analysis.

We firstly give case studies on aspect term extraction. We choose two examples to illustrate the effectiveness of the proposed model compared with SPAN. As shown in Table XI, the first example is about the multi-term problem. There are two words in the aspect term “silicone material”, but the SPAN model only extracts the word “material”, ignore “silicone”. Meanwhile, since biaffine-based constrains the process of span detection, the proposed model can extract the correct aspect term. The second example is a complex review with more than one aspect terms.

The review is hard to understand, there is more than one aspect in the review, and the opinion of the author always charges. Therefore, SPAN cannot extract any aspect terms. However, based on the scope detection model, it is easy for the proposed model to extract all the aspect terms from the review.

We then give case studies on aspect-based sentiment classification. As shown in Table XII, we choose two examples to illustrate the effectiveness of the proposed model compared with TD-BERT. The first example is about the scope mismatch problem. The aspect term in the example is a pan, but the author discussed too many things in the original text. Meanwhile, based on the scope detection model, it is easy for our proposed model to detect the scope of the given aspect term, and predict the correct polarity toward the aspect term. The second example is a hard understanding review. There are too many aspects in the original review, and the opinion of the author always charges. However, based on generating a simple and easy understanding clause, it is easy for the proposed model to give a correct polarity toward the aspect term.

VI. CONCLUSION

In this study, we consider the importance of scope detection in aspect-based sentiment analysis, and propose a scope detection model for aspect-based sentiment analysis. In particular, we firstly propose a scope detection model to distinguish whether the words from the review text are relevant with the aspect term, and filter irrelevant and noise information. We further investigate a biaffine-based model to extract the aspect terms based on the scope. In addition, we generate a simplified clause based on the scope of aspect term, and predict the polarity based on the simplified clause. We believe that the simplified clause is much easier to understand than the original review text. The experimental results indicate the effectiveness of the proposed model, and also show the importance of scope detection for aspect-based sentiment analysis.

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